ANALYSTLAB AFRICA

Data Science Internship Programme

Heart Disease Prediction
Statistical Analysis Summary

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Project: Heart Disease Prediction
Dataset: Heart Disease Prediction Dataset

Prepared for the AnalystLab Africa Data Science Internship Programme
Week 3 Submission – Advanced Data Analysis, Statistical Testing & Feature Engineering

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1. Introduction

This report presents the Statistical Analysis Summary for the Heart Disease Prediction project, prepared as part of the AnalystLab Africa Data Science Internship Programme, Week 3: Advanced Data Analysis, Statistical Validation & Feature Engineering.

Week 2 produced a cleaned and structured dataset, and the advanced exploratory analysis in Week 3 identified several patterns that appeared to distinguish patients with heart disease from those without it. Six statistical tests and one correlation analysis were carried out to check whether these patterns were genuine, rather than something that could easily have arisen by chance.

Consistent with the Week 3 brief, the emphasis throughout this report is on what each result means and why it matters, rather than on the statistical mechanics behind it. Each section below states the question being asked, summarises the result in plain terms, explains what it means for patients and for the dataset, and sets out the practical implication for the business.

2. Approach to Statistical Testing

Three types of comparison were needed, and a different, well-established test was used for each:

- **Comparing a numeric measurement (Age, MaxHR, Oldpeak) between patients with and without heart disease:** the Mann-Whitney U test was used, since it compares two groups without assuming the data follows a normal distribution (Mann & Whitney, 1947).
- **Checking whether a category (ST_Slope, Exercise Angina, Chest Pain Type) is linked to heart disease:** the Chi-Square test of independence was used, the standard way of testing whether two categorical variables are related (Pearson, 1900).
- **Checking whether the numeric variables overlap with one another:** Spearman's rank correlation was used, which captures a general tendency for two variables to rise or fall together, even if the relationship isn't a straight line (Spearman, 1904).

Every test below returned a p-value below 0.001, which in practice means the patterns observed are very unlikely to be due to chance. Rather than repeating this statistical detail for every variable, the sections that follow focus on interpreting what each result actually tells us.

3. Age vs. Heart Disease

Research Question: Are patients with heart disease typically older than patients without it?

Result: Yes. Age was found to differ meaningfully between the two groups ($p < 0.001$).

What This Means: Patients with heart disease are, on average, significantly older than those without it. This confirms age as a crucial factor in heart disease risk.

Business Implication: Age can serve as an important initial screening factor. Healthcare providers should prioritize older individuals for closer monitoring and further cardiovascular assessments, although age should not be the sole determinant of risk.

4. Maximum Heart Rate (MaxHR) vs. Heart Disease

Research Question: Do patients with heart disease reach a lower maximum heart rate during exercise?

Result: Yes. MaxHR was found to differ meaningfully between the two groups ($p < 0.001$).

What This Means: Patients with heart disease tend to have a lower maximum heart rate compared to healthy individuals. Reduced MaxHR capacity is strongly associated with the presence of heart disease.

Business Implication: MaxHR is a valuable indicator for heart disease risk assessment. Integrating MaxHR into screening and predictive models can help identify patients needing additional evaluation, supporting targeted resource allocation and earlier intervention.

5. Oldpeak vs. Heart Disease

Research Question: Is greater ST depression during exercise (Oldpeak) linked to heart disease?

Result: Yes. Oldpeak was found to differ meaningfully between the two groups ($p < 0.001$).

What This Means: Higher Oldpeak values (indicating greater ST depression during exercise) are significantly associated with heart disease. The heart disease group also shows greater variability in Oldpeak.

Business Implication: Oldpeak is an important diagnostic indicator. Including Oldpeak in risk assessment can help healthcare professionals identify high-risk patients who require further investigation, leading to earlier intervention.

6. ST_Slope vs. Heart Disease

Research Question: Is the shape of the ST segment during exercise (ST_Slope) linked to heart disease?

Result: Yes. A strong association was found between ST_Slope and heart disease status ($p < 0.001$).

What This Means: The pattern of ST segment slope (downsloping, flat, or upsloping) is not independent of heart disease status. Specific ST_Slope patterns (like flat or downsloping) are notably more prevalent in patients with heart disease.

Business Implication: ST_Slope characteristics are highly useful in assessing a patient's likelihood of heart disease. Clinicians should consider these patterns alongside other indicators for robust risk assessment and screening tools.

7. Exercise Angina vs. Heart Disease

Research Question: Are patients who experience chest pain during exercise more likely to have heart disease?

Result: Yes. A significant association was found between exercise-induced angina and heart disease ($p < 0.001$).

What This Means: The presence or absence of exercise-induced angina is strongly related to heart disease status. Patients experiencing angina during exercise are significantly more likely to have heart disease.

Business Implication: Exercise-induced angina provides crucial information for identifying individuals at higher risk. This can guide clinical assessment and follow-up strategies.

8. Chest Pain Type vs. Heart Disease

Research Question: Does the type of chest pain a patient reports relate to their likelihood of heart disease?

Result: Yes. Chest pain type was found to be significantly associated with heart disease ($p < 0.001$).

What This Means: The type of chest pain reported (e.g., asymptomatic, atypical angina, non-anginal pain, typical angina) is not independent of heart disease status. Certain chest pain types (especially asymptomatic) show a very high prevalence of heart disease.

Business Implication: Chest pain type is a valuable diagnostic factor. Different chest pain categories may necessitate varying levels of clinical attention and investigation to accurately assess heart disease risk.

9. Correlation Analysis

Research Question: Do the numeric clinical measurements (Age, RestingBP, Cholesterol, MaxHR, Oldpeak) largely overlap with one another, or does each one add distinct information?

Result: Most numerical variables show weak to very weak relationships with one another. The strongest is a moderate negative relationship between Age and MaxHR ($r = -0.365$).

What This Means: The numerical health indicators largely provide distinct pieces of information, with no evidence of severe overlap (multicollinearity) among them.

Business Implication: Since these variables are not highly correlated, they can effectively complement each other in a heart disease risk-prediction model, offering unique insights without redundant information. This is beneficial for building robust predictive models.

10. Overall Business Takeaways

- **Key Predictors:** Age, MaxHR, Oldpeak, ST_Slope, Exercise Angina, and Chest Pain Type are all statistically significant indicators for heart disease. These factors should be prioritized in risk assessment and diagnostic pathways.
- **Integrated Risk Assessment:** Due to the interplay of multiple factors, a holistic approach combining these variables is crucial for accurate risk stratification rather than relying on any single indicator.
- **Targeted Interventions:** Understanding the specific associations can help healthcare systems design more targeted screening programs and personalized prevention strategies, leading to earlier diagnosis and improved patient outcomes.

11. Conclusion

Taken together, these results tell a consistent story: age, exercise capacity (MaxHR), exercise-induced ST depression (Oldpeak), ST slope pattern, exercise-induced angina, and chest pain type are all meaningfully linked to heart disease in this dataset, while the numeric measurements themselves do not simply duplicate one another.

The practical value of this lies less in the statistics themselves and more in what they confirm: a small set of clinical characteristics, considered together rather than in isolation, offer a genuinely informative picture of a patient's heart disease risk. This gives confidence that these variables are worth carrying forward into feature selection and, ultimately, into the predictive model built in Week 4.

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